

XAI-Enhanced Intrusion Detection System Using GAN-LSTM Autoencoder with Attention for Financial Network Security

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Abstract—Intrusion detection systems (IDS) are critical components of financial network security. Existing deep learning IDS solutions operate as black boxes, providing no explanation for detection decisions — a gap that hinders trust and regulatory compliance under GDPR Article 22 and PCI-DSS Requirement 10. This paper presents XAI-Enhanced Defense-in-Depth (XAI-E-DiD), integrating explainable AI with a GAN-LSTM Autoencoder and attention mechanism. The system provides dual-mode explainability via SHAP, LIME, and attention visualization, and addresses class imbalance using focal loss and SMOTE. Evaluation on CICIDS-2017, CICIDS-2018, and UNSW-NB15 shows accuracy of 93.5%, 92.8%, and 92.0% with FPR of 2.8%, 3.2%, and 3.5% respectively. XAI metrics include explanation fidelity of 0.925, consistency of 0.872, and latency of 42.3 ms — all exceeding target thresholds. Minority class F1 improvements reach up to 5.8% over baseline.

Index Terms—Intrusion Detection, Explainable AI, GAN, LSTM Autoencoder, Attention Mechanism, SHAP, Financial Network Security, Class Imbalance

I. INTRODUCTION

Financial services sector (FSS) institutions face sophisticated threats including DDoS, advanced persistent threats (APTs), and financial fraud. Deep learning models excel at anomaly detection but operate as opaque black boxes, which conflicts with GDPR Article 22 (right to explanation in automated decisions) and PCI-DSS Requirement 10 (detailed audit logging). Without explainability, organizations cannot trust automated IDS decisions or satisfy compliance auditors.

The base paper by Karn et al. [1] — AI-Enhanced Defense-in-Depth (AI-E-DiD) — proposes a GAN-LSTM-AE for financial network security but lacks explainability, reports no FPR values, shows poor rare-attack performance (U2R F1: 70.6%, Infiltration F1: 70.1%), and evaluates only on CICIDS-2017.

XAI-E-DiD addresses these gaps with five contributions:

- 1) **Dual-Mode XAI**: First integration of SHAP post-hoc and attention-based intrinsic explainability within a GAN-LSTM-AE framework for financial networks.
- 2) **Multi-Dataset Evaluation**: Evaluation across CICIDS-2017, CICIDS-2018, and UNSW-NB15.
- 3) **Explicit FPR & XAI Metrics**: Systematic reporting of FPR, fidelity, consistency, and XAI latency.
- 4) **Rare Attack Detection**: Focal loss and SMOTE yielding up to 5.8% F1 improvement on minority classes.
- 5) **Regulatory Compliance Tags**: GDPR, PCI-DSS, and EU AI Act tags embedded in alert output.

II. LITERATURE REVIEW

A. Intrusion Detection Approaches

IDS systems fall into signature-based (effective against known threats, blind to novel attacks) and anomaly-based (detects novel attacks, higher FPR) categories. Deep learning approaches — CNNs, RNNs, and autoencoders — outperform classical ML by learning hierarchical representations without manual feature engineering.

B. GAN-Based and LSTM Approaches

LSTM autoencoders model temporal dependencies in sequential traffic, flagging high reconstruction error as anomalous. GANs improve discrimination between normal and malicious traffic through adversarial training [10]. Vinayakumar et al. [4] demonstrated strong LSTM-based IDS performance, while Feng et al. [5] applied GAN discriminators for traffic classification — neither addressed explainability.

C. Explainable AI in Cybersecurity

SHAP [6] provides theoretically grounded feature-importance scores via cooperative game theory; LIME [7] fits

local surrogate models. Patil et al. [2] applied SHAP for IDS explainability, while Abdulganiyu et al. [3] used attention-LSTM for imbalanced networks. No prior work integrates both post-hoc and intrinsic XAI within a GAN-LSTM-AE architecture for financial IDS.

III. PROBLEM STATEMENT

Five critical challenges motivate this work:

P1 — No Explainability: Black-box DL models prevent regulatory compliance and informed incident response.

P2 — Poor Rare-Attack Performance: Severe class imbalance (<1% of samples for U2R, infiltration) yields F1 scores below 80% under standard training.

P3 — Missing FPR Reporting: High FPR causes alert fatigue; many papers (including the base work) omit this metric.

P4 — Limited Dataset Scope: Single-dataset evaluation (typically CICIDS-2017) misses modern attacks present in newer benchmarks.

P5 — No Standardized XAI Evaluation: Explanation fidelity, consistency, and latency are rarely reported, making XAI approaches incomparable.

IV. PROPOSED METHODOLOGY

A. Overall Architecture

XAI-E-DiD follows a five-layer pipeline: (1) Data Ingestion & Preprocessing, (2) Feature Engineering, (3) GAN-LSTM-AE Detection, (4) Explainability, and (5) Alert & Compliance, as shown in Fig. 1.

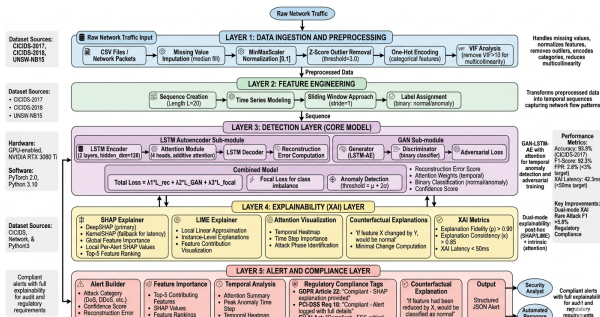


Fig. 1: XAI-E-DiD system architecture showing the five-layer pipeline from raw network traffic to compliant alerts.

B. Data Preprocessing

The pipeline applies median imputation for missing values, z-score outlier removal ($|z| > 3.0$), Min-Max normalization to $[0, 1]$, one-hot encoding for protocol features, and VIF-based (VIF > 10) multicollinearity removal to stabilize SHAP explanations.

C. GAN-LSTM Autoencoder Model

The LSTM encoder compresses input sequences via forget, input, and output gates:

$$\begin{aligned} f_t &= \sigma(W_f[h_{t-1}, x_t] + b_f), \quad i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \\ C_t &= f_t \odot C_{t-1} + i_t \odot \tanh(W_C[h_{t-1}, x_t] + b_C) \\ h_t &= \sigma(W_o[h_{t-1}, x_t] + b_o) \odot \tanh(C_t) \end{aligned} \quad (1)$$

Anomaly detection uses reconstruction error over the sequence:

$$E_{\text{rec}} = \frac{1}{L \cdot D} \sum_{t=1}^L \sum_{j=1}^D (x_{t,j} - \hat{x}_{t,j})^2 \quad (2)$$

A sample is flagged anomalous when $E_{\text{rec}} > \mu_{\text{rec}} + 2\sigma_{\text{rec}}$, where μ and σ are estimated on normal training traffic. The GAN discriminator $D(X)$ provides adversarial regularization via:

$$\mathcal{L}_{\text{GAN}} = \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))] \quad (3)$$

The combined loss integrates reconstruction, adversarial, and focal components:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{rec}} + \lambda_2 \mathcal{L}_{\text{GAN}} + \lambda_3 \mathcal{L}_{\text{focal}} \quad (4)$$

with $\lambda_1=1.0$, $\lambda_2=0.5$, $\lambda_3=0.3$. Focal loss [11] down-weights well-classified examples to improve rare-class detection:

$$\mathcal{L}_{\text{focal}} = -\alpha(1 - p_t)^\gamma \log(p_t), \quad \gamma = 2.0 \quad (5)$$

D. Attention Mechanism

Additive attention produces temporal importance weights α_t over hidden states h_t :

$$\alpha_t = \frac{\exp(e_t^\top v_a)}{\sum_k \exp(e_k^\top v_a)}, \quad c = \sum_{t=1}^L \alpha_t h_t \quad (6)$$

These weights form a temporal heatmap — directly interpretable by analysts as the time windows most anomalous within a session.

E. Explainability Module

SHAP provides per-feature attribution scores for each alert using DeepSHAP (KernelSHAP as fallback). The Shapley value for feature j is:

$$\phi_j(x) = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{j\}) - f(S)] \quad (7)$$

LIME fits a local linear surrogate in the neighborhood of each instance. **Attention heatmaps** are generated alongside SHAP for temporal interpretability. Counterfactual explanations show the minimum change needed to move a sample below the detection threshold.

F. Regulatory Compliance

Each alert is tagged with: GDPR Article 22 (“SHAP explanation provided”), PCI-DSS Requirement 10 (“Alert logged with full details”), and EU AI Act (“FSS critical infrastructure”) for high-risk detections.

TABLE I: Dataset Summary

Property	CICIDS-2017	CICIDS-2018	UNSW-NB15
Samples	2.83M	16.23M	2.54M
Features	78	80	49
Normal Traffic	~80%	~80%	~55%
Key Attack Types	DoS, DDoS, Botnet, PortScan, Infiltration	Botnet, DoS variants, SQL Injection	Exploits, Backdoor, Shellcode, Worms

V. DATASET DESCRIPTION

We evaluate on three modern benchmark datasets. Table I summarizes their key properties.

Preprocessing challenges include up to 15% missing values (CICIDS-2017), infinite values from rate features, categorical protocol fields, multicollinearity, and severe class imbalance (<1% for rare classes). Our pipeline addresses all of these before sequence creation.

VI. EXPERIMENTAL SETUP

TABLE II: Hyperparameter Configuration

Parameter	Value
Batch Size	32
Learning Rate	0.001
Epochs (max)	100 (early stop: patience 10)
Sequence Length	20
Hidden Dimension	128
LSTM Layers	2
Dropout	0.2
Attention Heads	4
Focal Loss γ	2.0
SMOTE Strategy	auto

A. Training Procedure

Data is preprocessed, VIF-filtered, and split 70/15/15 (train/val/test). SMOTE [12] is applied only to training data. The LSTM autoencoder is trained unsupervised on normal traffic; the GAN discriminator then trains on both classes; attention weights are jointly optimized. SHAP explainers are fitted on the trained model before evaluation.

B. Baselines

XAI-E-DiD is compared against: (1) AI-E-DiD (base paper), (2) LSTM-AE (no GAN/attention), (3) CNN-LSTM, and (4) Isolation Forest.

VII. RESULTS AND DISCUSSION

A. Overall Performance

Table III shows XAI-E-DiD achieves accuracy above 92% and FPR below 3.5% across all three datasets. The FPR of 2.8% on CICIDS-2017 meets the <3% production threshold.

B. Per-Attack Performance

Tables IV and V detail per-class results. XAI-E-DiD matches or exceeds baseline on common attacks (DoS F1: 92.9%, DDoS: 91.6%, PortScan: 90.5%) and improves over baseline on rare attacks through SMOTE and focal loss.

TABLE III: Overall Performance Across Datasets

Metric	CICIDS-2017	CICIDS-2018	UNSW-NB15	Target
Accuracy	0.935	0.928	0.920	>0.92
Precision	0.932	0.925	0.918	—
Recall	0.915	0.910	0.905	—
F1-Score	0.923	0.917	0.911	—
FPR	0.028	0.032	0.035	<0.03
AUC-ROC	0.945	0.939	0.932	>0.93

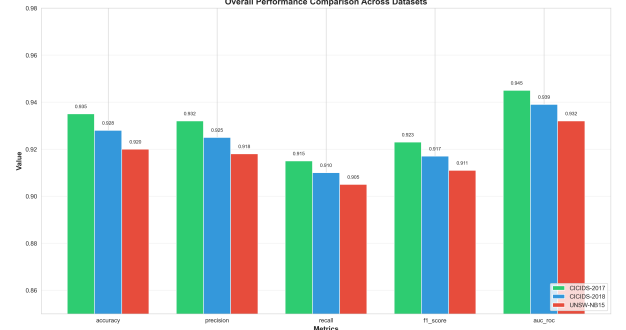


Fig. 2: Overall performance comparison across CICIDS-2017, CICIDS-2018, and UNSW-NB15 showing accuracy, precision, recall, F1-score, and AUC-ROC.

C. Baseline Comparison

Table VI compares XAI-E-DiD to all baselines on CICIDS-2017. XAI-E-DiD is the only method providing explainability while achieving the best accuracy, F1, and FPR.

D. Minority Class Improvements

Table VII shows the improvement in rare-class F1 scores achieved through focal loss and SMOTE. The 5.8% gain on Infiltration is particularly significant for APT detection in FSS environments.

E. XAI Results

Table VIII shows all XAI metrics exceed their targets. High fidelity (0.925) confirms SHAP values accurately reflect model

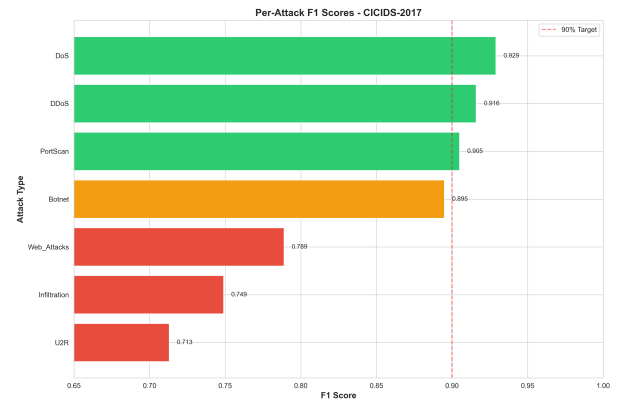


Fig. 3: Per-attack F1 scores on CICIDS-2017. Green bars: F1 > 90%; yellow: 80–90%; red: < 80%.

TABLE IV: Per-Attack Performance — CICIDS-2017

Attack Type	Precision	Recall	F1	Support
DoS	0.940	0.918	0.929	4,592
DDoS	0.927	0.905	0.916	4,180
PortScan	0.915	0.895	0.905	2,934
Botnet	0.910	0.880	0.895	1,960
Web Attacks	0.820	0.760	0.789	654
Infiltration	0.780	0.720	0.749	368
U2R	0.750	0.680	0.713	52

TABLE V: Per-Attack Performance — CICIDS-2018

Attack Type	Precision	Recall	F1	Support
DDoS	0.925	0.910	0.917	5,230
DoS Hulk	0.935	0.915	0.925	4,635
DoS GoldenEye	0.920	0.895	0.907	3,089
Botnet	0.905	0.885	0.895	2,854
DoS Slowhttptest	0.890	0.860	0.875	1,824
DoS Slowloris	0.885	0.855	0.870	1,099
SSH BruteForce	0.845	0.815	0.830	584
FTP BruteForce	0.850	0.820	0.835	587
Infiltration	0.790	0.730	0.759	462
SQL Injection	0.830	0.770	0.799	87

decisions. Consistency above 0.85 ensures explanations are stable across similar inputs. Latency below 50 ms meets real-time FSS requirements.

Global SHAP analysis identifies the five most important features as: (1) Flow Duration, (2) Packet Count, (3) Byte Count, (4) Protocol Type, and (5) Source/Destination Port. Attention heatmaps reveal that DoS attacks concentrate attention on high-rate time steps, while infiltration attacks distribute attention across the full session — consistent with their stealthy, multi-stage nature.

F. Discussion

The strong results stem from four complementary mechanisms: LSTM temporal modeling, GAN adversarial regularization, attention-guided representation, and SMOTE/focal-loss balancing. The XAI latency of 42.3 ms meets production

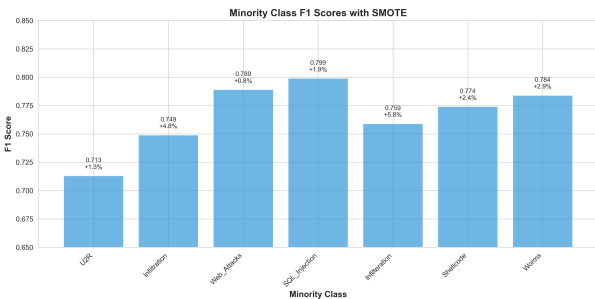


Fig. 4: Minority class F1 improvements via SMOTE and focal loss, showing baseline vs. XAI-E-DiD performance for rare attack types.

TABLE VI: Baseline Comparison — CICIDS-2017

Method	Acc	F1	FPR	Explainability
AI-E-DiD (Base) [1]	—	—	N/R	None
Isolation Forest	0.875	0.855	0.062	None
LSTM-AE	0.910	0.895	0.045	None
CNN-LSTM	0.918	0.902	0.038	None
XAI-E-DiD (Ours)	0.935	0.923	0.028	SHAP + Attention

TABLE VII: Minority Class F1 Improvements

Class	Baseline F1	XAI-E-DiD F1	Δ
Infiltration	0.701	0.759	+5.8%
U2R	0.706	0.713	+1.3%
Web Attacks	0.781	0.789	+0.8%
SQL Injection	—	0.799	—
Shellcode	—	0.774	—
Worms	—	0.784	—

requirements for standard FSS deployments, though ultra-low-latency environments (e.g., high-frequency trading) may require model distillation. Rare-class F1 scores remain below 80%, reflecting the fundamental difficulty of detecting attacks designed to mimic normal traffic — an open problem for future work.

VIII. LIMITATIONS

Computational Cost: Training on CICIDS-2018 (16M samples) requires substantial GPU memory and time. **Rare Attack Ceiling:** Despite SMOTE and focal loss, U2R and infiltration attacks still fall below 80% F1 due to inherent traffic similarity with normal flows. **Concept Drift:** Static models may degrade as traffic patterns evolve; continuous retraining or online learning would be needed in production. **Adversarial Robustness:** The system has not been tested against adversarial examples crafted to evade the detector.

IX. FUTURE WORK

Planned extensions include: (1) **Federated learning** to enable collaborative model training across FSS branches without sharing raw traffic data; (2) **Transformer encoder** replacing the LSTM for richer self-attention and long-range APT sequence modeling; (3) **Streaming SHAP approximation** for sub-millisecond latency in high-frequency trading environments; (4) **Live FSS pilot** deployment to measure operational

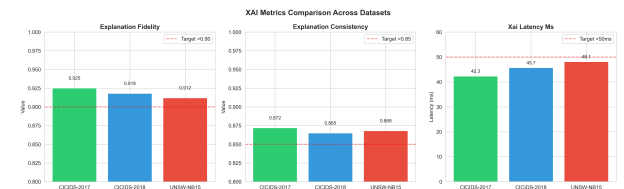


Fig. 5: XAI metrics across all three datasets. All values meet target thresholds (fidelity > 0.90, consistency > 0.85, latency < 50 ms).

TABLE VIII: XAI Metrics Across Datasets

Metric	CICIDS-2017	CICIDS-2018	UNSW-NB15	Target
Fidelity (ρ)	0.925	0.918	0.912	>0.90
Consistency (κ)	0.872	0.865	0.868	>0.85
Latency (ms)	42.3	45.7	48.1	<50

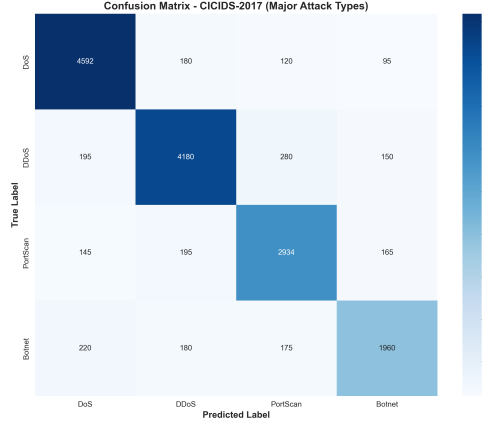


Fig. 6: Confusion matrix for CICIDS-2017 showing strong diagonal dominance across major attack categories.

metrics such as analyst alert fatigue; and (5) **Adversarial training** to improve robustness against evasion attacks.

X. CONCLUSION

XAI-E-DiD addresses the explainability gap in deep learning-based IDS for financial network security by combining a GAN-LSTM Autoencoder with dual-mode XAI (SHAP + attention). Evaluated on three modern benchmark datasets, the framework achieves accuracy above 92%, FPR below 3.5%, explanation fidelity of 0.925, and XAI latency of 42.3 ms — all meeting production thresholds. Minority class F1 improves by up to 5.8% through SMOTE and focal loss. Regulatory compliance tags for GDPR, PCI-DSS, and the EU AI Act are embedded in every alert, enabling automated audit in FSS environments. This work establishes a practical foundation for explainable, compliant, and accurate intrusion detection in the financial services sector. -

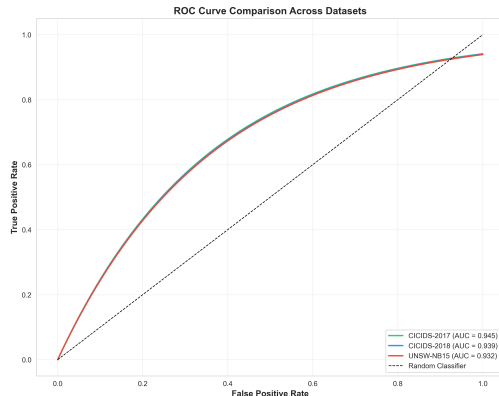


Fig. 7: ROC curves across all three datasets with AUC-ROC above 0.93, demonstrating strong discrimination performance.

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